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VTU.REG.NO

B.E. First Semester Examination – May 2023

Introduction to Electronics Engineering

(Common to CSE, CSD, ISE, and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) List the limitations of a half wave rectifier. Neatly draw the circuit diagram of a Bi-phase full wave rectifier. Describe the working of the circuit during positive half cycle and negative half cycle. Draw the waveforms one below another, properly aligned for the following : i) Transformer secondary ii) Current across diode D1 iii) Current across diode D2 iv) Output across load. **(10 Marks)**

   b) Why voltage regulator is required? Draw the circuit diagram for a zener diode shunt voltage regulator. Explain how the circuit will regulate the power supply at the output. Develop the expressions to find the minimum and maximum values for the current limiting resistor in the regulator circuit. **(10 Marks)**

2. a) Draw the circuit diagram of an amplifier using NPN transistor and explain its working with the help of the output characteristics (graph of collector current verses collector to emitter voltage). **(10 Marks)**

   b) Explain the working of BJT as a switch. Mention an operation mode of BJT when it is working as a switch in on and off condition. Draw relevant sketches for the purpose of explanation. **(10 Marks)**

Module – II

3. a) List the desirable characteristics for an ideal operational amplifier. Compare operational amplifier parameters for ideal and real devices with typical values. **(10 Marks)**

   b) Sketch the circuit for integrator using operational amplifier. Write the equation for output voltage. Draw the input and output waveforms one below another aligned properly. Consider input as square wave. **(10 Marks)**

4. a) Sketch neatly the block diagram showing an amplifier with positive feedback applied. Mention the input, output and feedback network in the sketch. Write the equation for overall gain interms of amplifier gain Av and feedback ratio (β). Explain the effect of positive feedback. **(10 Marks)**

   b) Sketch the sine wave oscillator band on a 3-stage CR ladder network. Explain how the different parts of the circuitry satisfy the conditions for oscillations. Write expression for frequency of oscillation. **(10 Marks)**

Module – III

5. a) Convert the following numbers from the given base to the other three bases indicated.
   i) Decimal 225 to binary, octal and hexadecimal.
   ii) Octal 623 to decimal, binary and hexadecimal. **(10 Marks)**

   b) Perform the subtraction with the following unsigned binary numbers by taking 2's complement of the subtrahand.
   i) 11010 – 10000 ii) 100 – 110000 **(10 Marks)**

6. a) Simplify the following Boolean expressions to the minimum number of literals
   i) $x' + xy + xz' + xy'z'$ *ii)* $(BC' + A'D)(AB + CD')$ **(10 Marks)**

   b) Obtain the truth table of the following function and express the function in sum of minterms and product of maxterms.
   $F(x, yz) = (xy + z)(y + xz)$ **(10 Marks)**

Module – IV

7. a) What is an embedded system? Critically compare embedded system and general computing system. **(10 Marks)**

   b) Classify the embedded system based on:
   i) Generation ii) Complexity and performance **(10 Marks)**

8. a) What are RISC and CISC processors / controllers? Compare critically. **(10 Marks)**

   b) What are sensors? How sensors are classified? Provide details of 5 common types of sensors. **(10 Marks)**

Module – V

9. a) Draw the block diagram representing schematic of an analog transmitter section and explain. **(10 Marks)**

   b) What is multiplexing in communication? Mention different types of multiplexing and explain. Also mention the benefits of multiplexing. **(10 Marks)**

10. a) Identify different types of analog modulation methods. Define each one of them and draw the waveforms (modulating signal, carrier signal, modulated signal) for each. **(10 Marks)**

    b) Explain the following briefly with relevant sketches.
    i) Ground wave propagation ii) Space wave propagation iii) Sky wave propagation. **(10 Marks)**